

Ostrom and the eVSM

Where a theory of collective-action institutions and a model of any viable institution overlap, and what each supplies that the other lacks

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The question

Ostrom's model, as Eric Jones redrew it in 2003, is about what lets a collective-action institution come into being, last, and perform. The eVSM (extended Viable System Model) is about what any institution must do to stay viable. If both are right, they should overlap heavily. This note sets out where, using two sources: Jones's figure as redrawn in the RCN Graph Tool (*Ostroms Collective Action in 3 phases*) and Marc's eVSM *plus environment for RCN*.

The short answer: the overlap is large. It sits mostly in Ostrom's **Survival** phase, partly in **Performance**, and there is almost none in **Origin**. The VSM assumes the system already exists; Origin is about whether it forms at all.

The overall layout already matches

Ostrom / Jones	eVSM
Left panel: the users and the resource	The operating units (OU-1 ... OU-n, the Operating Unit Hub) and their local environments
Center panel: the institution's practicalities	The metasystem: spheres S-1 to S-11
Right panel: external factors	The environment column (Regulators, Partners, Competitors, Financiers, Ideal Futures ...)

Jones puts Common Understanding and Decision-making Costs in the *users'* panel, not the institution's. In a commons the users do their own steering. Beer says the same: the metasystem is a set of functions, not a separate class of people. Ostrom's design principle 3 (the people affected by the rules help make them) is what that looks like in practice.

Three positions settled in discussion

1. Trust, Similarities, Experience and Open Communication live on the eVSM's edges

Matching nodes to nodes, these four looked weakly covered. They are not properties of any sphere; they are properties of the channels. Every sphere-to-sphere link in the eVSM runs both ways (33 of the 55 possible sphere pairs are linked, 50 two-way pairs in all), and the quality of those two-way links is where Ostrom's social variables sit:

- **Experience** is Ostrom's "mutually beneficial transactions over time": the history of a two-way channel settling into balance.
- **Trust** reduces how much each side must check the other. A trusted channel does not need auditing.

- **Open Communication** is how much a channel can carry.
- **Similarities** is how little translation a channel needs where two parts meet (Beer's transducers).

RCN's own instruments already read these edges: the Agree / Disagree / Unknown edge ratings and the Block G "who you work with" network in the eVSM measures specification.

The specific channels that carry them between users are the **informal OU-to-OU links**, and that is where most of the variety is handled. The current eVSM drawing links OU-1 to OU-2 and OU-2 to OU-n but not OU-1 to OU-n. Marc will redraw it with an informal channel between every pair of operating units.

Two limits. **Asset Structure** is about how holdings are divided, not about a relationship, so it belongs with Resources & Accounting (S-7), in who pays and who receives. And **Origin** comes before the edges exist: Ostrom's left panel describes how channels get built; the VSM describes channels that are working.

2. Coordination and monitoring: combine the watching, not the sanctioning

Ostrom supplies the argument for folding monitoring into Coordination (S-1). In working commons the monitors are the users themselves or answer to them (design principle 4). That makes monitoring a sideways, peer activity with the same shape as coordination. Jones's own causal chain runs the same way: Info on Others' Resource Use → Common Understanding ↔ Conflict Resolution.

The condition is to combine the *watching*, not the *sanctioning*. Sanctions stay with Managing (S-6) and Policies & Legal (S-10). Beer kept coordination (System 2) and audit (System 3*) apart for a reason: operations share information freely with a coordinator who serves them, and stop once the coordinator is also the inspector. If peers collude and nobody reports, Ostrom's backstop is Dual Enforcement, with Regulators outside.

Beer describes two levels of coordination: operating units coordinating directly with each other, and above them a coordination function that System 3 runs. That second level is an **S-1 to S-6 link**, which the current drawing does not have. It is needed; how to draw it is still open.

Open: how to show the inspection function without losing the simplicity, symmetry and readability of the eVSM metasystem. The candidates are adding edges rather than a node (for example return edges from each operating unit to the Hub, which are one-way in the current file), marking the watching as a property of S-1's existing edges, or a small marker on S-1. Nothing is chosen yet.

3. Ostrom's three levels map onto the eVSM's color bands

Ostrom level	eVSM spheres
Constitutional	S-10 Policies & Legal, S-11 Worldview & Identity (magenta)
Bridge	S-8 Planning, S-9 Research (purple): System 4, in both levels
Collective choice	S-5 Innovation, S-6 Managing, S-7 Resources & Accounting (red): System 3, where the budget is made, "taxes" collected and redistributed
Operational	The operating units, S-1 Coordination (teal), and the shared services S-2 HR, S-3 Infrastructure / Tools, S-4 Outreach (blue), which bridge System 3 and the operating units

- **Each band writes the rules for the band below.** In Ostrom, operational rules are made in the collective-choice arena, so S-6 and S-7 make them. The constitutional band decides who sits in the collective-choice arena and how its budget rules can change.
- **Why System 4 sits in both:** constitutional change is usually set off by what Research sees, a changing environment or collective choices that keep failing. That is Jones's Rate of Change → Survival link. Planning/Research also sits in a balancing pair with Managing, and that pair is itself a collective-choice arena.
- **Recursion:** every operating unit has its own metasystem and its own three levels. The larger system's collective-choice rules limit the local system's constitutional choices. That is Ostrom's nesting seen through Beer's recursion.

The Markov-blanket reading

Marc reads the eVSM as a Markov blanket that minimizes free energy (Friston). The sensory / motor states are S-4, S-7 and S-9; the tuning layer is S-1, S-2, S-5, S-10 and S-11; S-3, S-6 and S-8 are the internal core, the layer the eVSM introduction already calls the "internal core layer." Every operating unit is a smaller blanket of its own, one level of recursion down. The drawings mark each node's blanket role.

The blanket holds in the current file: every environment node connects only to S-4, S-7 or S-9. That is also why Regulators should reach Policies & Legal through Research rather than directly. Internal states may not touch the outside.

Ostrom fits this reading closely:

- **Common Understanding**, in Jones's words "information storage with a comparator inside it," is a prior plus a prediction-error signal, the core of free-energy minimization.
- **Ostrom's revised rule**, "the smallest and simplest set of rules relevant to the variety in the environment," is Ashby's Law of Requisite Variety, and it is free energy's trade-off between complexity and accuracy: the simplest model that still fits the world.
- **Rate of Change → Survival** is a world that changes faster than the model can update, so prediction error cannot be brought down.
- **Jones's claim that outside factors reach origin and performance only indirectly** is the blanket condition. His government arrows cross the boundary at a few points: Legal Status, de facto vs. de jure, Dual Enforcement, Technology for Monitoring and Speed of Access to Info. Those are Ostrom's sensory states. Where Jones draws government reaching into the users' panel, the blanket reading says the arrow lands on a sensory state first.

Node by node

strong same function or condition · partial overlapping · **channel** carried by the informal OU-to-OU edges, not by a sphere

Ostrom node	eVSM counterpart		Why
Outside			
Governmental Agencies	Regulators	strong	The most organized outside actor; reaches the eVSM through Research (S-9).
Exogenous Changes	Competitors, Market Geographies	partial	Shocks arrive through the environment column.
Rate of Change	S-9 Research	strong	Beer defines viability as adapting as fast as the environment changes.
Larger System of Institutional Support	The next recursion up; Partners	strong	Nested enterprises (design principle 8) are Beer's recursion.
External Factors	The environment column as a whole	partial	A heading, not a single node.
Constitutional			
Recognized Membership	S-11 Worldview & Identity	strong	Clear boundaries (design principle 1): who is inside, where the blanket is.
Common Understanding	S-11; also S-9, S-8	strong	Store plus comparator: the shared model, prior plus error.
Legal Status, Local Prerogative	S-10 Policies & Legal	strong	Legal standing, handled with Regulators through S-9.
de facto vs. de jure	S-10 Policies & Legal	strong	Whether the rules in use are the rules the state recognizes.
Bridge			
Perceived Alternative Strategies	S-8 Planning, S-5 Innovation	partial	The options the group can see, and the ones it can invent.
Info about Resource	S-9 Research, S-7 Resources & Accounting	partial	Sensing the resource; S-7 as measurement.
Collective choice			
Small, Simple Set of Rules	S-6 Managing; bounded by S-10	strong	Requisite variety. The strongest single match, the hinge between the two models.
Enforcement Costs	S-6 Managing; S-7	strong	Sanctioning is decided here and paid from the budget.
Decision-making Costs	S-6 Managing	strong	The cost of running the collective-choice arena. Group Size raises it: more variety for the metasystem to absorb.

Asset Structure	S-7 Resources & Accounting	strong	Who pays and who receives: the budget and its redistribution.
Dual Enforcement	S-6 Managing, S-10 Policies & Legal	partial	Local enforcement backed from outside; the backstop when peer monitoring fails.
Operational: functions			
Conflict Resolution	S-1 Coordination	strong	Cheap, routine dispute handling is System 2: damping oscillation between units.
Info on Others' Resource Use	S-1 Coordination; the OU-to-OU channels	strong	Peer monitoring: the watching, not the sanctioning.
Technology for Monitoring Resources	S-3 Infrastructure / Tools	strong	Instruments that make the resource observable are shared tools.
Speed of Access to Info	S-3; S-1	partial	The delay, not the data, is the variable.
Operational: the users			
Trust, Open Communication, Similarities, Experience	The informal OU-to-OU channels	channel	Properties of edges, not spheres (position 1 above).
Interdependence / subtractability	OU-to-OU channels; Suppliers (OU)	channel	Units drawing on the same resource have overlapping local environments, which is why they oscillate and need coordination.
Group Size	Operating Unit Hub	partial	How many operating units the metasystem must absorb.
Distance to Resource, Resource Scarcity	Suppliers (OU)	partial	The operating units' shared local environment.
Outcomes			
Survival	Viability	strong	The same concept.
Performance	Viability	partial	What viability buys; Beer measures it with actuality, capability and potentiality.
Origin	None		The VSM assumes the system exists.

What each supplies the other

What Ostrom gives the eVSM

- **Origin:** how the channels get built before they can carry anything.
- **Equity:** Asset Structure puts who gains and who pays into the budget question at S-7.
- **The state as a source of legitimacy,** not just another item in the environment. In the eVSM it arrives through S-9.
- **Dual enforcement** as the backstop behind peer monitoring.

What the eVSM gives Ostrom

- **Balancing the present against the future** (Beer's System 3–System 4 pair). Jones's map has nothing that plans or innovates; Perceived Alternatives is its only forward-looking node.
- **Separating coordination from management.** Ostrom merges them into "rules plus enforcement."
- **Alarm signals** (Beer's algedonic channel) that go straight to the top. Ostrom's map has no fast path for alarms.
- **Recursion stated outright.** Ostrom implies it with a single node.
- **Blanket discipline:** every outside influence lands on a sensory state first.

Changes to the eVSM

- **Agreed:** an informal channel between every pair of operating units (Marc will redraw).
- **Agreed:** an S-1 to S-6 link for Beer's second level of coordination. How to draw it is still to decide.
- **Agreed:** Coordination carries the watching; sanctions stay in S-6 and S-10.
- **Not needed:** a direct S-10 to Regulators edge. S-9 carries that traffic, as the blanket requires.
- **Open:** how to draw inspection without breaking the metasystem's symmetry.

The two drawings

Both are built in the RCN Graph Tool; the JSON files open there for editing. Hovering any node or line in the tool, or in the SVG files, shows the reasoning behind it.

- Crosswalk, strong match
- - - Crosswalk, partial match
- • Ostrom variable carried by the informal OU-to-OU channels (lines into the dot marked T01)
- - - eVSM change agreed in this discussion
- eVSM channel, as drawn
- ☐ dashed Added for the crosswalk: Next recursion up, Viability

Banded by level (next two pages): Ostrom on the left, the eVSM on the right, one band per level. It gives up the eVSM's geometric shape so the crosswalk lines never cross a node. The sphere-to-sphere mesh and environment couplings are left out; they are in the as-drawn view. The small italic line on each node gives its blanket role.

As drawn (the landscape page): both source diagrams with their own positions and links unchanged, joined by the crosswalk. It is busy on purpose: it shows where each source puts things. Each node's small italic lines give its IAD level and blanket role.

OSTROM / JONES 2003: conditions

eVSM FOR RCN: functions and environment

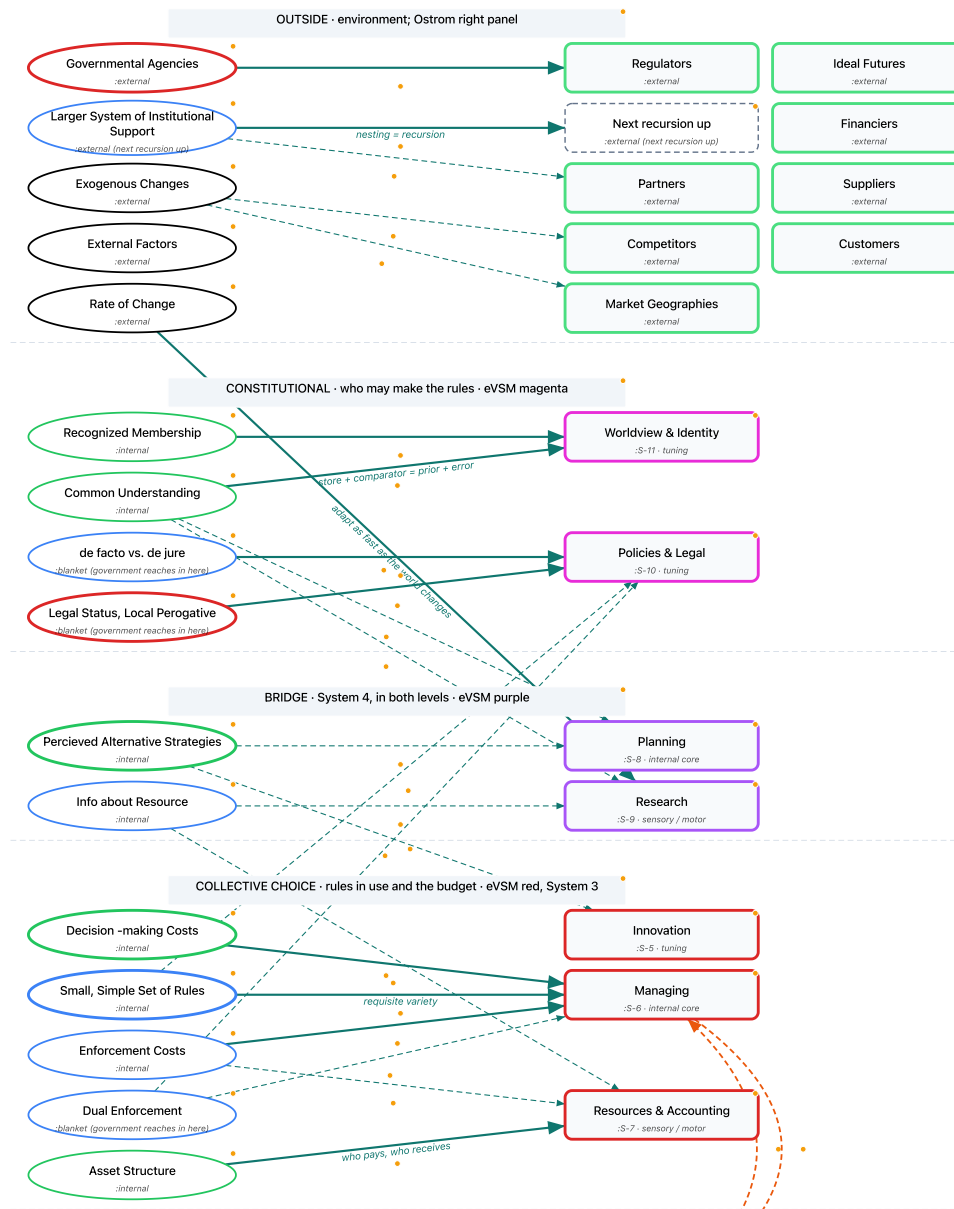


Figure 1a. Banded crosswalk: outside, constitutional, bridge and collective choice.

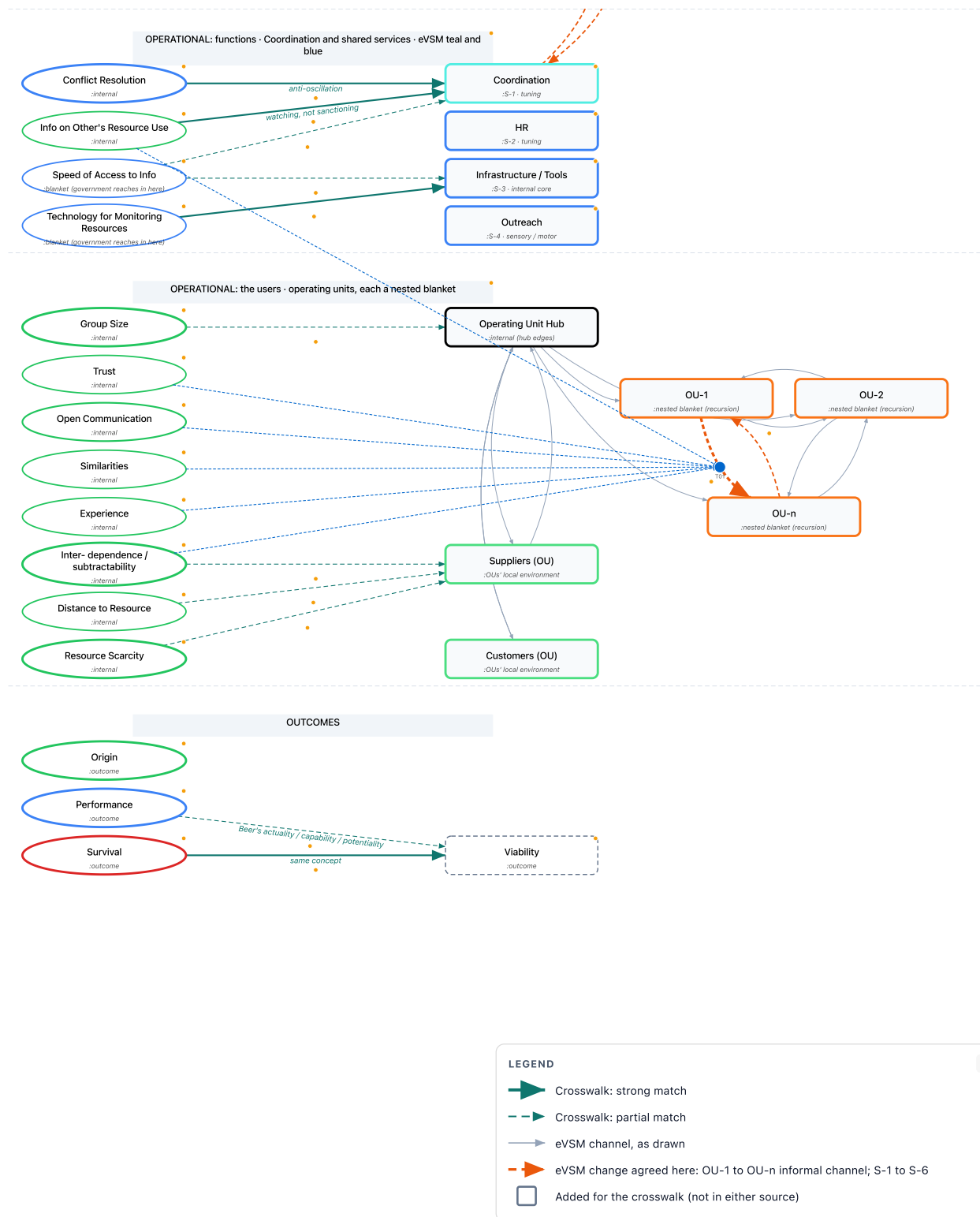


Figure 1b. Banded crosswalk: operational functions, the users and the outcomes. The orange S-1 to S-6 link continues from Figure 1a.

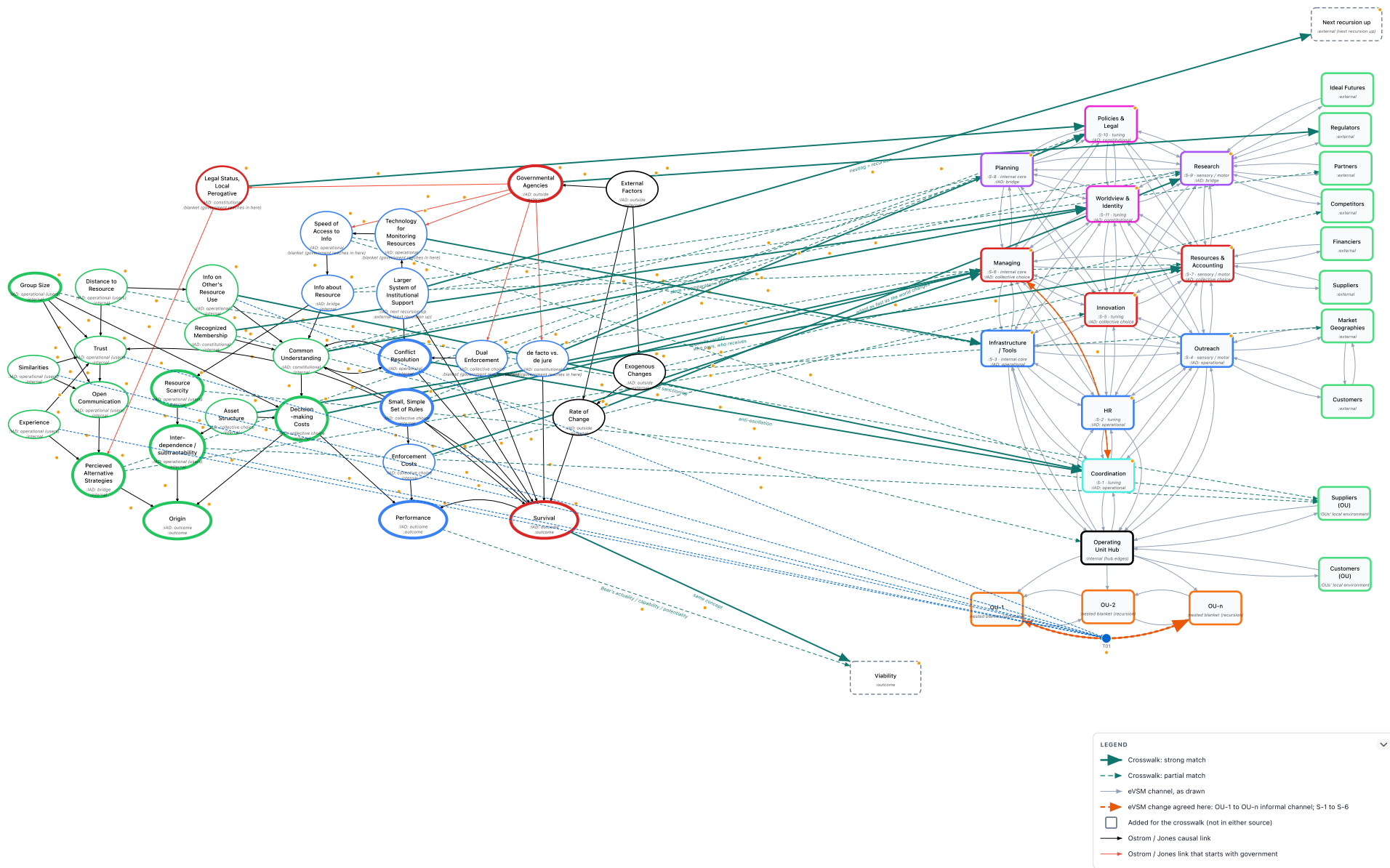


Figure 2. The two diagrams as drawn: Jones's redrawing of Ostrom (left), the eVSM plus environment (right), joined by the crosswalk.

Sources and files

- Jones, E. C. (2003). Building on Ostrom's "The Rudiments of a Theory of the Origins, Survival, and Performance of Common-property Institutions." *Journal of Ecological Anthropology* 7(1): 65–72, Figure 1. doi:10.5038/2162-4593.7.1.4
- Ostrom, E. (1990). *Governing the Commons*. Cambridge University Press (the design principles).
- Beer, S. (1979, 1985). *The Heart of Enterprise; Diagnosing the System for Organizations*.
- RCN: docs/evsm-intro.html, docs/evsm-measures-spec.html.
- Graph Tool files, in docs/ostrom-evsm/: ostrom-evsm-banded.json and ostrom-evsm-as-drawn.json (editable), each with .svg and .png exports.